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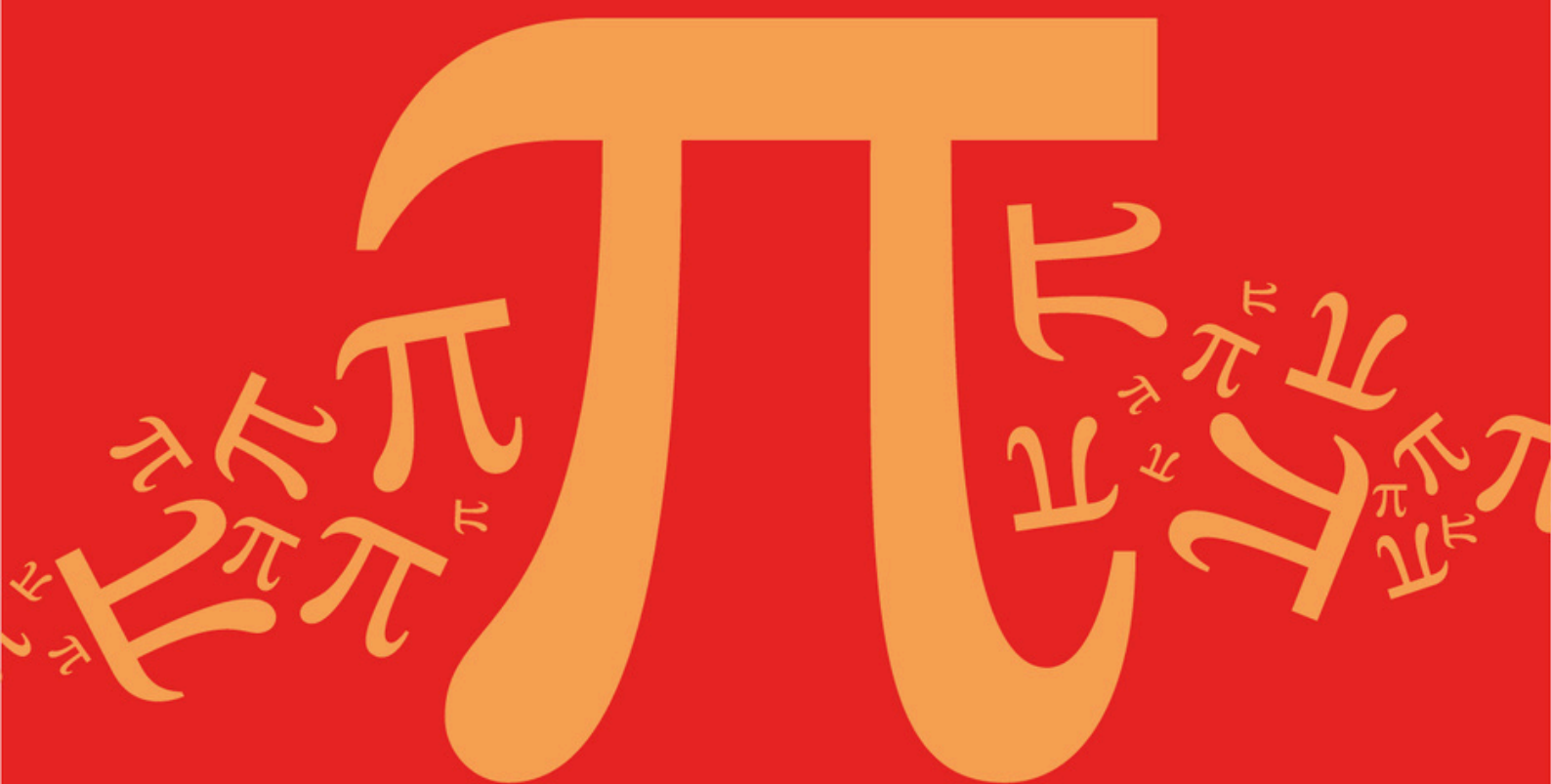
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AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D15: Graphs of Conics

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic **D: Further Algebra and Functions**
Sub topic **D15: Graphs of Conics**

EXAM PAPERS PRACTICE

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

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Name : _____

Date : _____

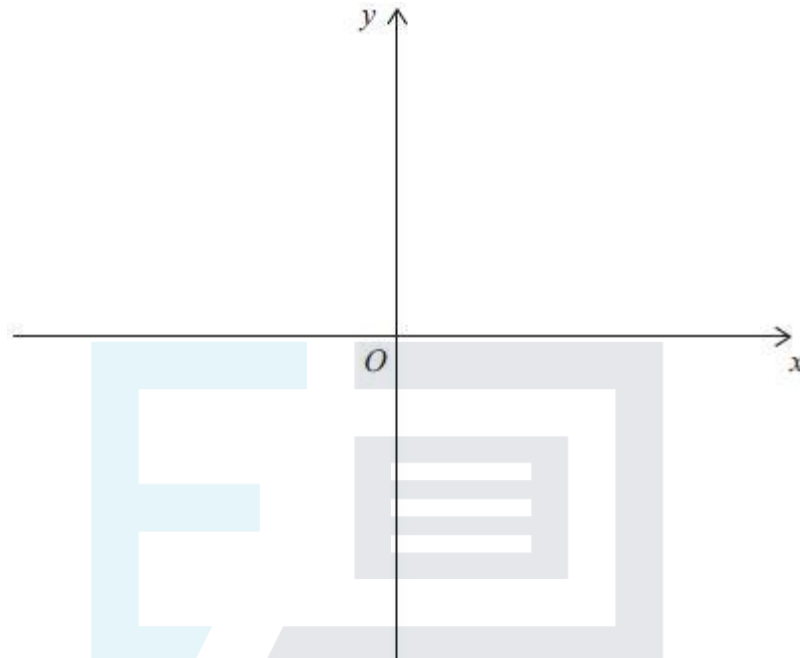
Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

- (a) Sketch the graph of $y^2 = 4x$



(1)

- (b) Ben is using a 3D printer to make a plastic bowl which holds exactly 1000 cm^3 of water.

Ben models the bowl as a region which is rotated through 2π radians about the x-axis.

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He uses the finite region enclosed by the lines $x = d$ and $y = 0$ and the curve with equation $y^2 = 4x$ for $y \geq 0$

- (i) Find the depth of the bowl to the nearest millimetre.

(4)

- (ii) What assumption has Ben made about the bowl?

(1)

(Total 6 marks)

Q2.

Find the equations of the asymptotes of the curve $x^2 - 3y^2 = 1$

Circle your answer.

$y = \pm 3x$

$y = \pm \frac{1}{3}x$

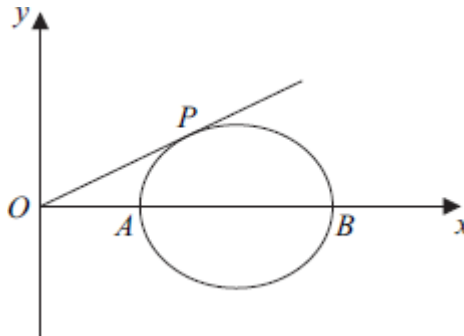
$y = \pm \sqrt{3}x$

$y = \pm \frac{1}{\sqrt{3}}x$

(Total 1 mark)

Q3.

An ellipse is shown below.



The ellipse intersects the x -axis at the points A and B . The equation of the ellipse is

$$\frac{(x-4)^2}{4} + y^2 = 1$$

- (a) Find the x -coordinates of A and B .

(2)

- (b) The line $y = mx$ ($m > 0$) is a tangent to the ellipse, with point of contact P .

- (i) Show that the x -coordinate of P satisfies the equation

$$(1 + 4m^2)x^2 - 8x + 12 = 0$$

(3)

- (ii) Hence find the exact value of m .

(4)

- (iii) Find the coordinates of P .

(4)

(Total 13 marks)

Q4.

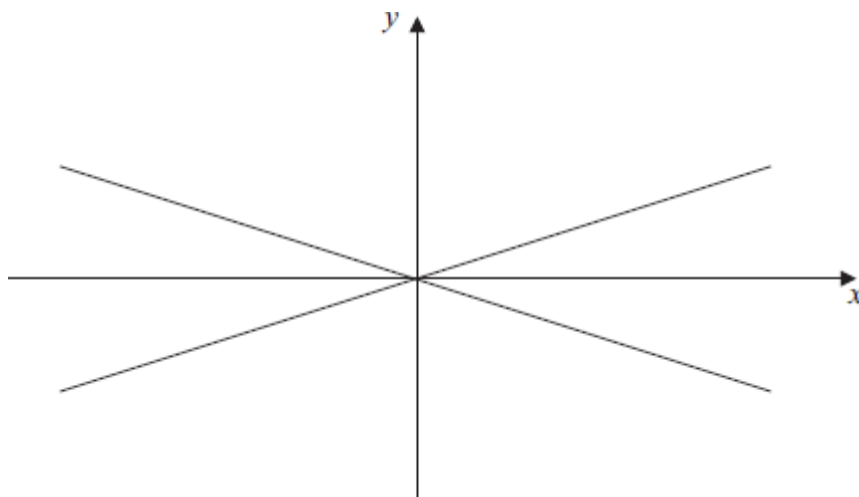
A hyperbola H has equation

$$\frac{x^2}{9} - y^2 = 1$$

- (a) Find the equations of the asymptotes of H .

(1)

- (b) The asymptotes of H are shown in the diagram below. On the same diagram, sketch the hyperbola H . Indicate on your sketch the coordinates of the points of intersection of H with the coordinate axes.



(3)

- (c) The hyperbola H is now translated by the vector $\begin{bmatrix} -3 \\ 0 \end{bmatrix}$.

(i) Write down the equation of the translated curve.

(2)

(ii) Calculate the coordinates of the two points of intersection of the translated curve with the line $y = x$.

(4)

- (d) From your answers to part (c)(ii), **deduce** the coordinates of the points of intersection of the original hyperbola H with the line $y = x - 3$.

(2)

(Total 12 marks)

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Q5.

The plane transformation T is a rotation through θ radians anticlockwise about O , and maps points (x, y) onto image points (X, Y) such that

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

where $c = \cos \theta$ and $s = \sin \theta$.

- (a) Write down the inverse of the matrix $\begin{bmatrix} c & -s \\ s & c \end{bmatrix}$ and hence show that

$$x = cX + sY \quad \text{and} \quad y = -sX + cY$$

(3)

- (b) The curve C has equation $x^2 - 6xy - 7y^2 = 8$.

The image of C under T is the curve C' with equation $pX^2 + qXY + rY^2 = 8$.

- (i) Use the results of part (a) to show that

$$q = 6s^2 + 16sc - 6c^2$$

and express p and r similarly in terms of c and s .

(4)

- (ii) Given that θ is an acute angle, find the values of c and s for which $q = 0$ and hence in this case express the equation of C' in the form

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

(8)

- (iii) Hence explain why C is a hyperbola.

(1)

(Total 16 marks)

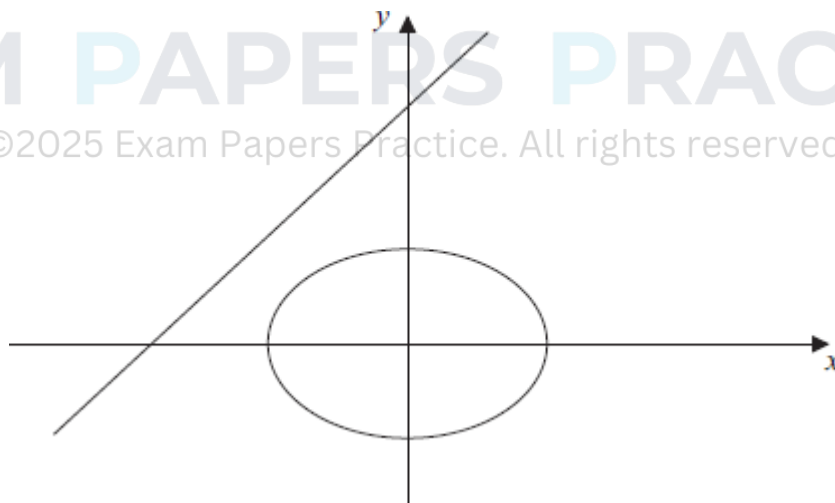
Q6.

The diagram shows the ellipse E with equation

$$\frac{x^2}{5} + \frac{y^2}{4} = 1$$

and the straight line L with equation

$$y = x + 4$$



- (a) Write down the coordinates of the points where the ellipse E intersects the coordinate axes.

(2)

- (b) The ellipse E is translated by the vector $\begin{bmatrix} p \\ 0 \end{bmatrix}$, where p is a constant. Write down the equation of the translated ellipse.

(2)

- (c) Show that, if the translated ellipse intersects the line L , the x -coordinates of the points of intersection must satisfy the equation

$$9x^2 - (8p - 40)x + (4p^2 + 60) = 0$$

(3)

- (d) Given that the line L is a tangent to the translated ellipse, find the coordinates of the two possible points of contact.

(No credit will be given for solutions based on differentiation.)

(8)

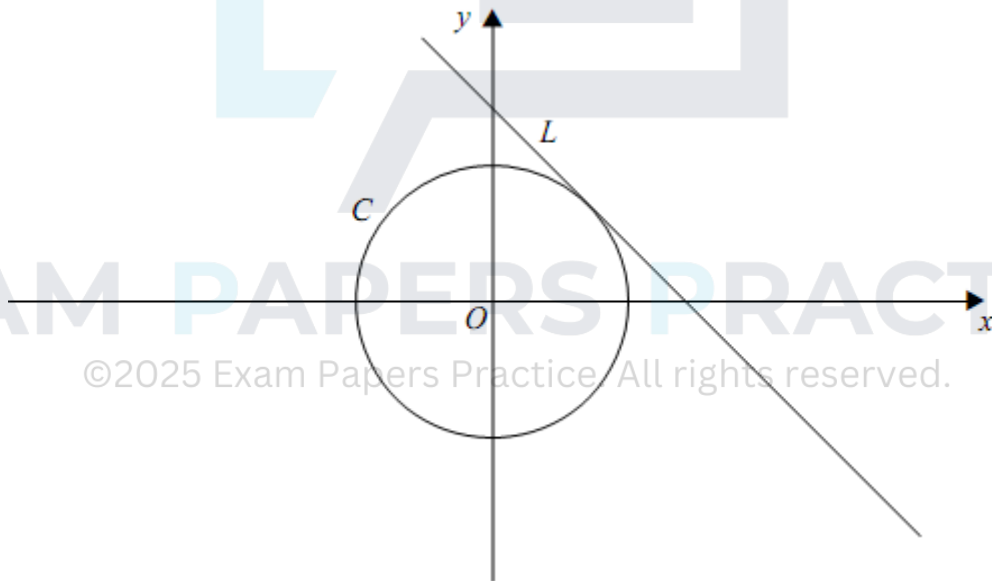
(Total 15 marks)

Q7.

The diagram shows a circle C and a line L , which is the tangent to C at the point $(1, 1)$. The equations of C and L are

$$x^2 + y^2 = 2 \text{ and } x + y = 2$$

respectively.



The circle C is now transformed by a stretch with scale factor 2 parallel to the x -axis. The image of C under this stretch is an ellipse E .

- (a) **On the diagram below**, sketch the ellipse E , indicating the coordinates of the points where it intersects the coordinate axes.

(4)

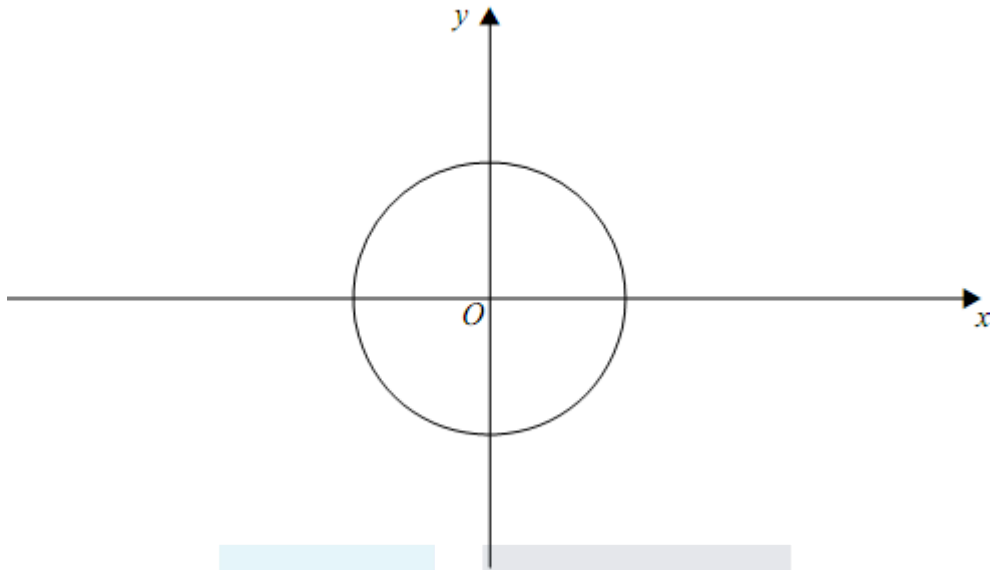
- (b) Find equations of:

(i) the ellipse E ;

(2)

(ii) the tangent to E at the point $(2, 1)$.

(2)



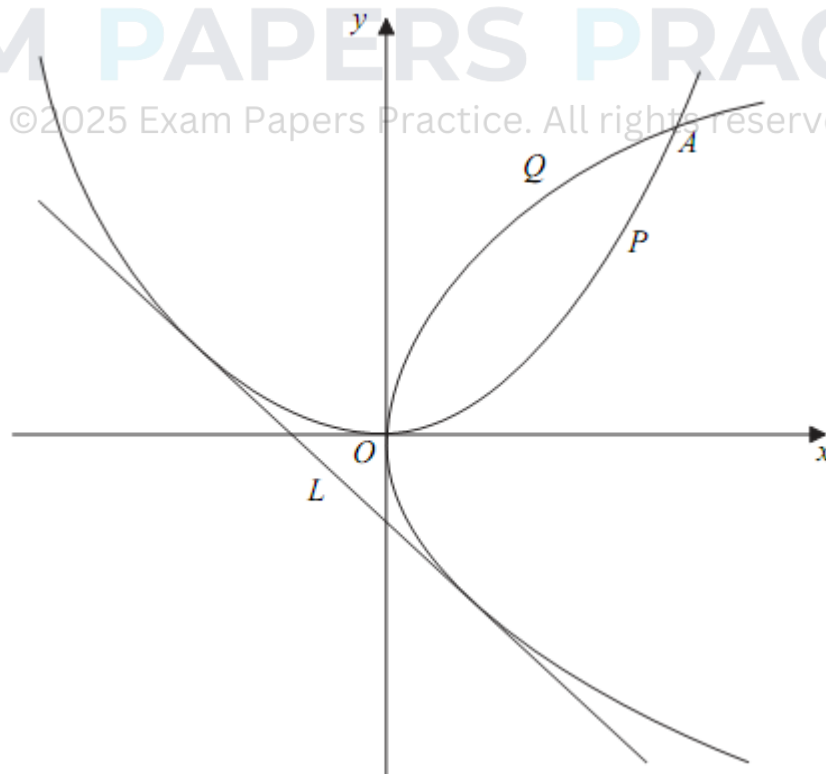
(Total 8 marks)

Q8.

The diagram shows a parabola P which has equation $y = \frac{1}{8}x^2$, and another parabola Q which is the image of P under a reflection in the line $y = x$.

The parabolas P and Q intersect at the origin and again at a point A .

The line L is a tangent to both P and Q .



- (a) (i) Find the coordinates of the point A .

(2)

(ii) Write down an equation for Q . (1)

(iii) Give a reason why the gradient of L must be -1 (1)

(b) (i) Given that the line $y = -x + c$ intersects the parabola P at two distinct points, show that

$$c > -2$$
 (3)

(ii) Find the coordinates of the points at which the line L touches the parabolas P and Q .

(No credit will be given for solutions based on differentiation.)

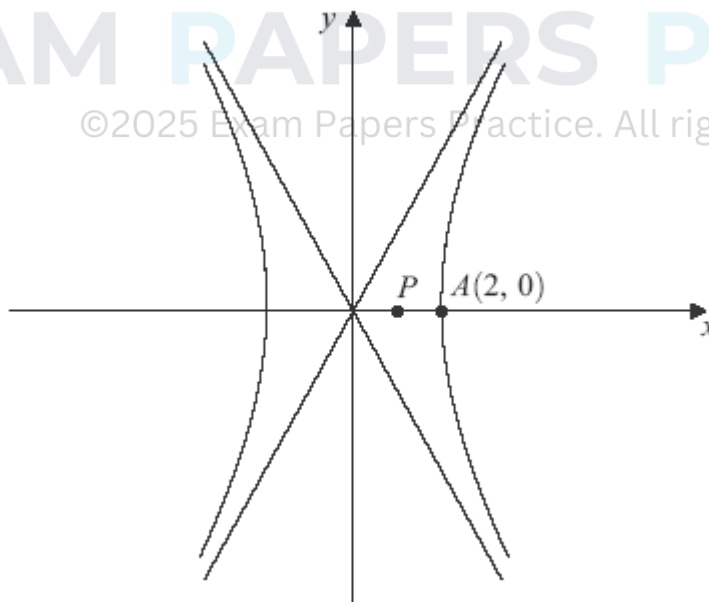
(4)
(Total 11 marks)

Q9.

The diagram shows the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

and its asymptotes.



The constants a and b are positive integers.

The point A on the hyperbola has coordinates $(2, 0)$.

The equations of the asymptotes are $y = 2x$ and $y = -2x$.

(a) Show that $a = 2$ and $b = 4$.

(4)

- (b) The point P has coordinates $(1, 0)$. A straight line passes through P and has gradient m . Show that, if this line intersects the hyperbola, the x -coordinates of the points of intersection satisfy the equation

$$(m^2 - 4)x^2 - 2m^2x + (m^2 + 16) = 0$$

(4)

- (c) Show that this equation has equal roots if $3m^2 = 16$.

(3)

- (d) There are two tangents to the hyperbola which pass through P . Find the coordinates of the points at which these tangents touch the hyperbola.

(No credit will be given for solutions based on differentiation.)

(5)

(Total 16 marks)

Q10.

A parabola P has equation $y^2 = x - 2$.

- (a) (i) Sketch the parabola P .

(2)

- (ii) On your sketch, draw the two tangents to P which pass through the point $(-2, 0)$.

(2)

- (b) (i) Show that, if the line $y = m(x + 2)$ intersects P , then the x -coordinates of the points of intersection must satisfy the equation

$$m^2x^2 + (4m^2 - 1)x + (4m^2 + 2) = 0$$

(3)

- (ii) Show that, if this equation has equal roots, then

$$16m^2 = 1$$

(3)

- (iii) Hence find the coordinates of the points at which the tangents to P from the point $(-2, 0)$ touch the parabola P .

(3)

(Total 13 marks)

Q11.

A hyperbola H has equation

$$x^2 - \frac{y^2}{2} = 1$$

- (a) Find the equations of the two asymptotes of H , giving each answer in the form $y = mx$.

(2)

- (b) Draw a sketch of the two asymptotes of H , using roughly equal scales on the two coordinate axes. Using the same axes, sketch the hyperbola H .

(3)

- (c) (i) Show that, if the line $y = x + c$ intersects H , the x -coordinates of the points of intersection must satisfy the equation

$$x^2 - 2cx - (c^2 + 2) = 0$$

(4)

- (ii) Hence show that the line $y = x + c$ intersects H in two distinct points, whatever the value of c .

(2)

- (iii) Find, in terms of c , the y -coordinates of these two points.

(3)

(Total 14 marks)

Q12.

The plane transformation T has matrix $\mathbf{A} = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$, and maps points (x, y) onto image points (X, Y) such that

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix}$$

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- (a) (i) Find $\mathbf{A}^{-1}$.

(2)

- (ii) Hence express each of x and y in terms of X and Y .

(2)

- (b) Give a full geometrical description of T .

(5)

- (c) Any plane curve with equation of the form $\frac{x^2}{p} + \frac{y^2}{q} = 1$, where p and q are distinct positive constants, is an ellipse.

- (i) Show that the curve E with equation $6x^2 + y^2 = 3$ is an ellipse.

(1)

- (ii) Deduce that the image of the curve E under T has equation

$$2X^2 + 4XY + 5Y^2 = 15$$

(2)

- (iii) Explain why the curve with equation $2x^2 + 4xy + 5y^2 = 15$ is an ellipse.

(1)

(Total 13 marks)

Q13.

An ellipse E has equation

$$\frac{x^2}{3} + \frac{y^2}{4} = 1$$

- (a) Sketch the ellipse E , showing the coordinates of the points of intersection of the ellipse with the coordinate axes.

(3)

- (b) The ellipse E is stretched with scale factor 2 parallel to the y -axis.

Find and simplify the equation of the curve after the stretch.

(3)

- (c) The **original** ellipse, E , is translated by the vector $\begin{bmatrix} a \\ b \end{bmatrix}$. The equation of the translated ellipse is

$$4x^2 + 3y^2 - 8x + 6y = 5$$

Find the values of a and b .

(5)

(Total 11 marks)

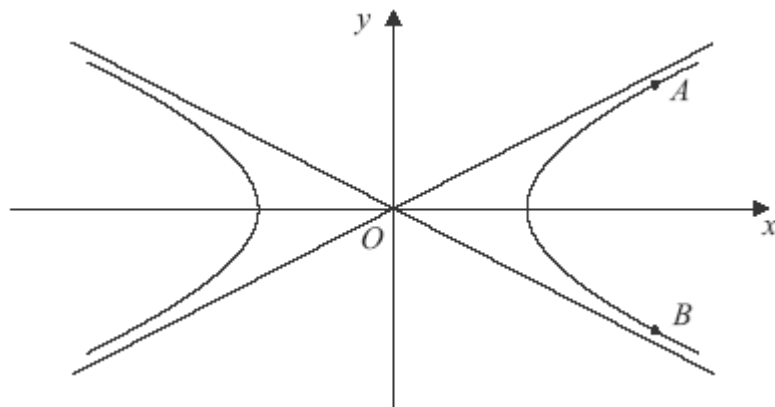
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Q14.

The diagram shows the hyperbola

$$\frac{x^2}{4} - y^2 = 1$$

and its asymptotes.



- (a) Write down the equations of the two asymptotes.

(2)

- (b) The points on the hyperbola for which $x = 4$ are denoted by A and B .

Find, in surd form, the y -coordinates of A and B .

(2)

- (c) The hyperbola and its asymptotes are translated by two units in the positive y direction.

Write down:

- (i) the y -coordinates of the image points of A and B under this translation;

(1)

- (ii) the equations of the hyperbola and the asymptotes after the translation.

(3)

(Total 8 marks)

Q15.

A curve C has equation

$$y = 7 + \frac{1}{x+1}$$

- (a) Define the translation which transforms the curve with equation $y = \frac{1}{x}$ onto the curve C .

(2)

- (b) (i) Write down the equations of the two asymptotes of C .

(2)

- (ii) Find the coordinates of the points where the curve C intersects the coordinate axes.

(3)

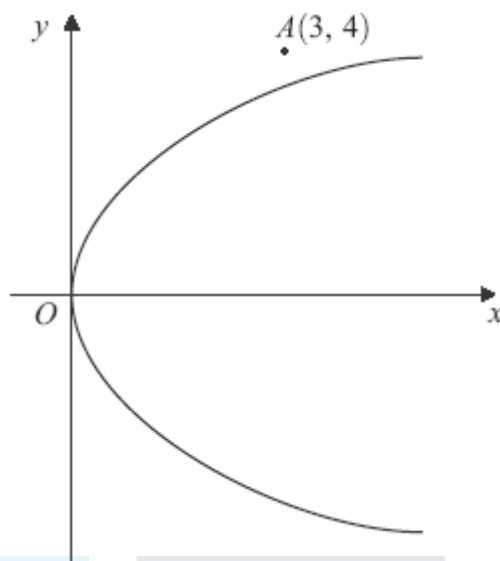
- (c) Sketch the curve C and its two asymptotes.

(3)

(Total 10 marks)

Q16.

The diagram shows the parabola $y^2 = 4x$ and the point A with coordinates $(3, 4)$.



- (a) Find an equation of the straight line having gradient m and passing through the point $A(3, 4)$. (2)
- (b) Show that, if this straight line intersects the parabola, then the y -coordinates of the points of intersection satisfy the equation (3)
- $$my^2 - 4y + (16 - 12m) = 0$$
- (c) By considering the discriminant of the equation in part (b), find the equations of the two tangents to the parabola which pass through A . (5)
 (No credit will be given for solutions based on differentiation.)
- (d) Find the coordinates of the points at which these tangents touch the parabola. (4)
- (Total 14 marks)**

Q17.

A curve C has equation

$$\frac{x^2}{25} - \frac{y^2}{9} = 1$$

- (a) Find the y -coordinates of the points on C for which $x = 10$, giving each answer in the form $k\sqrt{3}$, where k is an integer. (3)
- (b) Sketch the curve C , indicating the coordinates of any points where the curve intersects the coordinate axes. (3)
- (c) Write down the equation of the tangent to C at the point where C intersects the

positive x -axis.

(1)

- (d) (i) Show that, if the line $y = x - 4$ intersects C , the x -coordinates of the points of intersection must satisfy the equation

$$16x^2 - 200x + 625 = 0$$

(3)

- (ii) Solve this equation and hence state the relationship between the line $y = x - 4$ and the curve C .

(2)

(Total 12 marks)

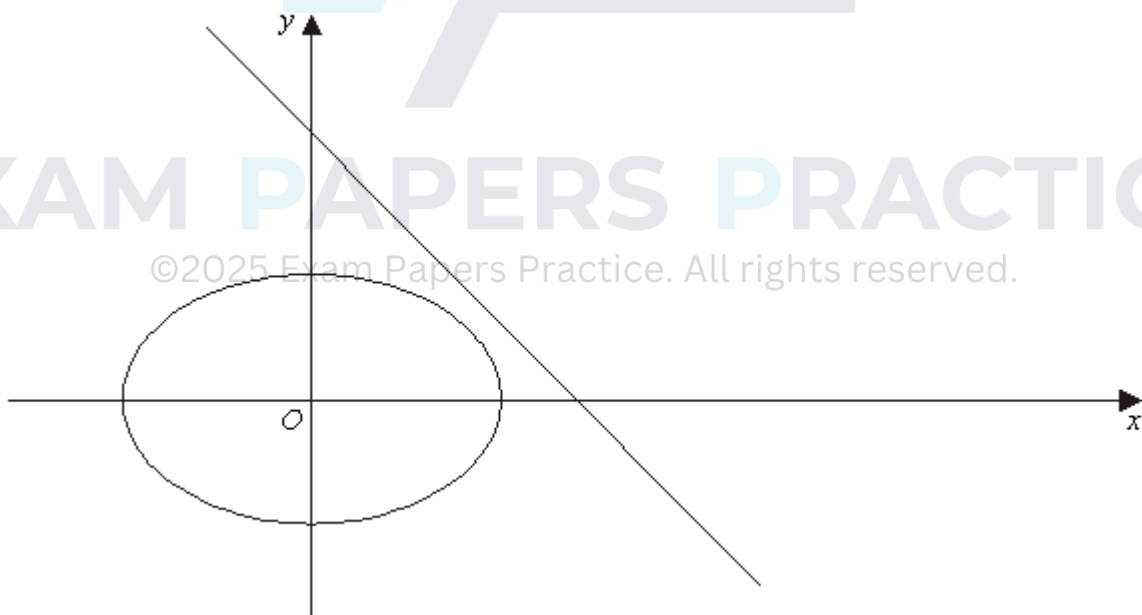
Q18.

The diagram below shows the curve with equation

$$\frac{x^2}{2} + y^2 = 1$$

and the straight line with equation

$$x + y = 2$$



- (a) Write down the exact coordinates of the points where the curve with equation $\frac{x^2}{2} + y^2 = 1$ intersects the coordinate axes.

(2)

- (b) The curve is translated k units in the positive x direction, where k is a constant. Write down, in terms of k , the equation of the curve after this translation.

(2)

- (c) Show that, if the line $x + y = 2$ intersects the **translated** curve, the x -coordinates of

the points of intersection must satisfy the equation

$$3x^2 - 2(k+4)x + (k^2 + 6) = 0$$

(4)

- (d) Hence find the two values of k for which the line $x + y = 2$ is a tangent to the translated curve. Give your answer in the form $p \pm \sqrt{q}$, where p and q are integers.

(4)

- (e) On the figure, show the translated curves corresponding to these two values of k .

(3)

(Total 15 marks)

Q19.

A curve has equation $y^2 = 12x$.

- (a) Sketch the curve.

(2)

- (b) (i) The curve is translated by 2 units in the positive y direction. Write down the equation of the curve after this translation.

(2)

- (ii) The **original** curve is reflected in the line $y = x$. Write down the equation of the curve after this reflection.

(1)

- (c) (i) Show that if the straight line $y = x + c$, where c is a constant, intersects the curve $y^2 = 12x$, then the x -coordinates of the points of intersection satisfy the equation

$$x^2 + (2c - 12)x + c^2 = 0$$

(3)

- (ii) Hence find the value of c for which the straight line is a tangent to the curve.

(2)

- (iii) Using this value of c , find the coordinates of the point where the line touches the curve.

(2)

- (iv) In the case where $c = 4$, determine whether the line intersects the curve or not.

(3)

(Total 15 marks)

Q20.

- (a) Describe a geometrical transformation by which the hyperbola

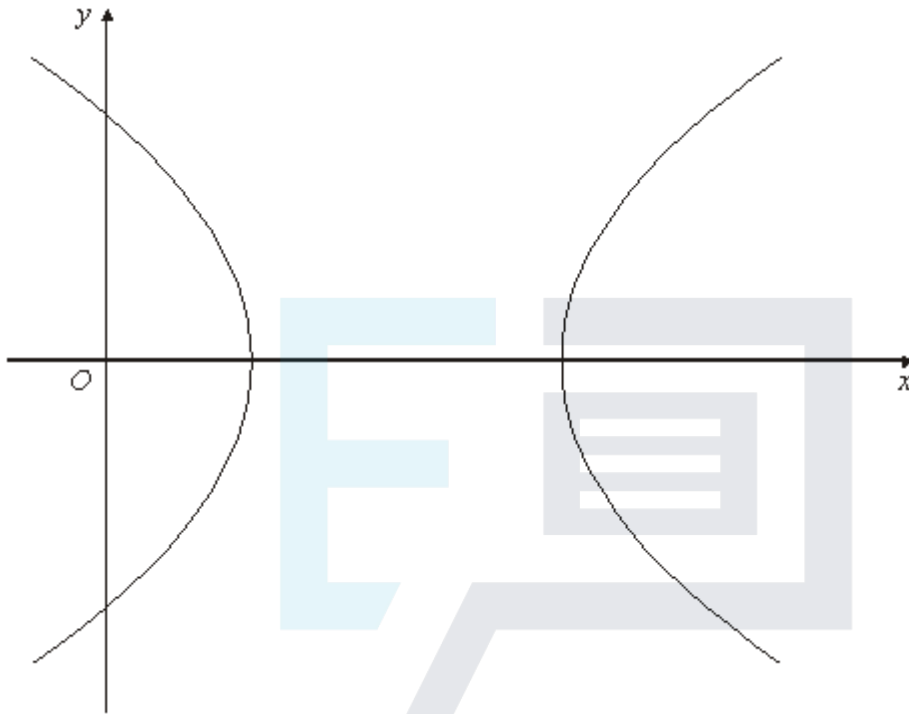
$$x^2 - 4y^2 = 1$$

can be obtained from hyperbola $x^2 - y^2 = 1$.

(2)

- (b) The diagram shows the hyperbola H with equation

$$x^2 - y^2 - 4x + 3 = 0$$



By completing the square, describe a geometrical transformation by which the hyperbola H can be obtained from hyperbola $x^2 - y^2 = 1$.

(4)

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Q21.

A curve has equation

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

- (a) Sketch the curve, showing the coordinates of the points of intersection with the coordinate axes.
- (b) Calculate the y -coordinates of the points of intersection of the curve with the line $x = 1$. Give your answers in the form $p\sqrt{2}$, where p is a rational number.
- (c) The curve is translated one unit in the positive x direction. Write down the equation of the curve after the translation.

(3)

(3)

(2)

(Total 8 marks)